

A Compactness Characterization of Weakly Random Sigma-Algebras

Autonomous solution packet for arXiv:1003.5588, Question 1

11 August 2026

Abstract

We answer Szegedy's question under the paper's standing separability assumption. A sub-sigma-algebra $\mathcal{B}$ of the product space is weakly random if and only if the conditional rank-one tensors

$$\mathbf{E}(f(x)\overline{g(y)} \mid \mathcal{B}), \quad \|f\|_\infty, \|g\|_\infty \leq 1,$$

form a relatively compact subset of $L^1(\mathcal{B})$. Equivalently, it is enough to test conditional expectations of rectangle indicators. Thus weak randomness is exactly compactness of a natural bilinear conditional-expectation map. One direction is a finite-net argument; the other follows from Arzelà–Ascoli on the weakly compact L^∞ unit ball in $L^2(\mathcal{B})$.

1 The question and the characterization

Throughout, $(V, \mathcal{A}, \mu)$ is the separable probability space fixed in the source paper, $X = V \times V$, and $P_{\mathcal{B}} = \mathbf{E}(\cdot \mid \mathcal{B})$ for a sub-sigma-algebra $\mathcal{B} \subseteq \mathcal{A} \otimes \mathcal{A}$. The source definition and Question 1 appear on PDF page 16 [1]:

Definition 1.5 *Let $(V, \mathcal{A}, \mu)$ be a probability space with σ -algebra $\mathcal{A}$ and measure μ . Let $\mathcal{B} \subseteq \mathcal{A} \times \mathcal{A}$ be a sub σ -algebra on the product space $V \times V$. We call $\mathcal{B}$ weakly random if in the set of functions $\{M \mid M \in L_\infty(\mathcal{B}), \|M\|_\infty \leq 1\}$ weak convergence implies cut norm convergence.*

Proposition 1.4 says that if V is the sphere S_2 and $\mathcal{B}$ consists of those Borel measurable sets that are invariant under the diagonal action of O_3 on $V \times V$ then $\mathcal{B}$ is weakly random.

Question 1 *Is there any characterization of weakly random σ -algebras?*

The Hilbert inner product is $\langle a, b \rangle = \int_X \overline{a}b \, d(\mu \otimes \mu)$, as in the source. Define

$$\begin{aligned} \mathcal{K}_{\mathcal{B}} &= \{P_{\mathcal{B}}(f \otimes \overline{g}) : f, g \in L^\infty(V), \|f\|_\infty, \|g\|_\infty \leq 1\}, \\ \mathcal{R}_{\mathcal{B}} &= \{P_{\mathcal{B}}(\mathbf{1}_S \otimes \mathbf{1}_T) : S, T \in \mathcal{A}\}. \end{aligned}$$

Both families lie in the unit ball of $L^\infty(\mathcal{B})$.

Theorem 1 (complete characterization). *The following are equivalent.*

- (i) $\mathcal{B}$ is weakly random in the sense of Szegedy: on $U_{\mathcal{B}} = \{M \in L^\infty(\mathcal{B}) : \|M\|_\infty \leq 1\}$, weak L^2 convergence implies cut-norm convergence.

(ii) $\mathcal{K}_{\mathcal{B}}$ is relatively compact in $L^1(\mathcal{B})$.

(iii) $\mathcal{R}_{\mathcal{B}}$ is relatively compact in $L^1(\mathcal{B})$.

Equivalently, the bounded bilinear map

$$\beta_{\mathcal{B}} : L^\infty(V) \times L^\infty(V) \longrightarrow L^1(\mathcal{B}), \quad \beta_{\mathcal{B}}(f, g) = P_{\mathcal{B}}(f \otimes \bar{g}),$$

is compact. In finite-net form, for every $\varepsilon > 0$ there are finitely many $h_1, \dots, h_m \in L^1(\mathcal{B})$ such that every conditional rectangle $P_{\mathcal{B}}(\mathbf{1}_S \otimes \mathbf{1}_T)$ is within ε of some h_j in L^1 .

2 Proof intuition

For a $\mathcal{B}$ -measurable kernel, conditional expectation moves every cut-norm test onto $\mathcal{B}$. Hence the cut norm is uniform evaluation against $\mathcal{K}_{\mathcal{B}}$. Uniform evaluation on a norm-precompact family is controlled by a finite net, which immediately converts weak convergence into cut convergence.

The converse uses the same identity in reverse. The L^∞ unit ball is a compact metrizable space in its weak L^2 topology. Weak randomness says that its cut-norm metric is continuous on this compact space. Therefore the linear functions indexed by $\mathcal{K}_{\mathcal{B}}$ are equicontinuous. Arzelà–Ascoli makes them compact in uniform norm, and L^1 – L^∞ duality identifies that uniform norm exactly with the L^1 distance between their indices.

3 The proof

For $M \in L^2(\mathcal{B})$, the definition of the source cut norm gives

$$\|M\|_{\square} = \sup_{\|f\|_{\infty}, \|g\|_{\infty} \leq 1} \left| \int_{V^2} \overline{f(x)} M(x, y) g(y) d\mu(x) d\mu(y) \right| = \sup_{k \in \mathcal{K}_{\mathcal{B}}} |\langle k, M \rangle|. \quad (1)$$

Indeed, the unprojected tensor in the middle pairing is $f \otimes \bar{g}$, and orthogonal projection onto $L^2(\mathcal{B})$ does not change its pairing with the $\mathcal{B}$ -measurable M .

Proof that (ii) implies (i). Suppose $M_n, M \in U_{\mathcal{B}}$ and $M_n \rightharpoonup M$ in $L^2(\mathcal{B})$. Given $\varepsilon > 0$, choose $k_1, \dots, k_m \in \mathcal{K}_{\mathcal{B}}$ so every $k \in \mathcal{K}_{\mathcal{B}}$ satisfies $\|k - k_j\|_1 < \varepsilon$ for some j . For such a pair,

$$|\langle k, M_n - M \rangle| \leq |\langle k_j, M_n - M \rangle| + \|k - k_j\|_1 \|M_n - M\|_{\infty} < |\langle k_j, M_n - M \rangle| + 2\varepsilon.$$

The maximum of the finitely many first terms tends to zero by weak convergence. Taking the supremum in (1), then $\varepsilon \downarrow 0$, proves $\|M_n - M\|_{\square} \rightarrow 0$. $\square$

Proof that (i) implies (ii). Regard $U_{\mathcal{B}}$ as a subset of the Hilbert space $H = L^2(\mathcal{B})$. It is convex, norm closed, and bounded, hence weakly compact. The standing separability assumption makes H separable, so the weak topology on $U_{\mathcal{B}}$ is metrizable.

Let $d_{\square}(M, N) = \|M - N\|_{\square}$. Assumption (i) says precisely that the identity

$$\iota : (U_{\mathcal{B}}, \text{weak } L^2) \longrightarrow (U_{\mathcal{B}}, d_{\square})$$

is sequentially continuous. Its domain is metrizable, so ι is continuous. For $k \in \mathcal{K}_{\mathcal{B}}$ define $\phi_k(M) = \langle k, M \rangle$. Each ϕ_k is weakly continuous, and

$$|\phi_k(M) - \phi_k(N)| \leq d_{\square}(M, N).$$

Continuity of ι therefore makes the family $\{\phi_k : k \in \mathcal{K}_{\mathcal{B}}\}$ equicontinuous on the compact space $(U_{\mathcal{B}}, \text{weak } L^2)$. It is also pointwise bounded. Arzelà–Ascoli shows that this family is relatively compact in $C(U_{\mathcal{B}})$ with the uniform norm.

For $k, \ell \in \mathcal{K}_{\mathcal{B}}$, $k - \ell$ is $\mathcal{B}$ -measurable, and the measurable phase of $k - \ell$ belongs to $U_{\mathcal{B}}$. Consequently

$$\|\phi_k - \phi_\ell\|_{C(U_{\mathcal{B}})} = \sup_{N \in U_{\mathcal{B}}} |\langle k - \ell, N \rangle| = \|k - \ell\|_1. \quad (2)$$

Thus $k \mapsto \phi_k$ is an isometry from $\mathcal{K}_{\mathcal{B}}$ with its L^1 metric into $C(U_{\mathcal{B}})$. Relative compactness of the image proves (ii). $\square$

Equivalence of (ii) and (iii). The implication (ii) $\Rightarrow$ (iii) is immediate because $\mathcal{R}_{\mathcal{B}} \subseteq \mathcal{K}_{\mathcal{B}}$. Conversely, suppose $\mathcal{R}_{\mathcal{B}}$ is L^1 -precompact. If $0 \leq u, v \leq 1$, the layer-cake formula gives

$$P_{\mathcal{B}}(u \otimes v) = \int_0^1 \int_0^1 P_{\mathcal{B}}(\mathbf{1}_{\{u>s\}} \otimes \mathbf{1}_{\{v>t\}}) ds dt,$$

where the integral is in L^1 . Hence this function belongs to the closed convex hull of $\mathcal{R}_{\mathcal{B}}$. The closed convex hull of a norm-precompact set in a Banach space is compact: replace the set by a finite ε -net and observe that the convex hull of that finite net is compact.

Decomposing the real and imaginary parts of each unit-bounded f, g into positive and negative parts expresses $P_{\mathcal{B}}(f \otimes \bar{g})$ as a sum of at most sixteen fixed scalar multiples of members of that compact convex hull. A finite sum of compact sets is compact, so $\mathcal{K}_{\mathcal{B}}$ is precompact. This proves (iii) $\Rightarrow$ (ii) and completes the proof of Theorem 1. $\square$

4 Sanity checks and scope

If $\mathcal{B}$ is finite, then $L^1(\mathcal{B})$ is finite-dimensional and bounded subsets are precompact, so every finite sub-sigma-algebra is weakly random. This agrees with the sequential definition.

At the opposite extreme, let V be an atomless standard probability space and $\mathcal{B} = \mathcal{A} \otimes \mathcal{A}$. Choose Rademacher functions r_n on V and put $M_n(x, y) = r_n(x)$. The sequence is orthonormal, hence weakly null in $L^2(V^2)$, but taking $f = r_n$ and $g = 1$ gives $\|M_n\|_{\square} = 1$. Thus the full product sigma-algebra is not weakly random. Equivalently, the associated conditional rank-one tensors are not L^1 -precompact.

The separability hypothesis is not cosmetic: it converts the sequential condition in the source into ordinary continuity on $U_{\mathcal{B}}$. For a nonseparable probability space, Theorem 1 remains true if weak randomness is formulated for weakly convergent nets; the source question, however, lives under its initial separability assumption and uses sequences.

The result is a functional-analytic characterization, not a classification up to measure-space isomorphism. It gives a concrete finite- L^1 -net test and exactly identifies the compact operator property behind the definition. Exact-phrease and run-index searches found no explicit later answer; this is only a novelty check, not part of the proof.

5 Verification

The proof was separately audited at the following pressure points: conditional-expectation conjugations in (1), weak compactness and metrizability of $U_{\mathcal{B}}$, the passage from sequential to ordinary continuity, complex L^1 duality in (2), and the rectangle layer-cake reduction. The finite verifier checks (1) and (2) over randomized finite product partitions; those computations are sanity checks, not substitutes for the proof.

References

- [1] B. Szegedy, *Limits of kernel operators and the spectral regularity lemma*, arXiv:1003.5588v1 (2010), Definition 1.5 and Question 1, p. 16.